

Scientific Machine Learning: Blending Science with Data

Homework Day 3

Consider the function below

$$f(x) = x \exp(-x^2)$$

- a) Plot $f(x)$ and comment on how many local maximum and minimum there are.
- b) Use the optimization packages in Julia to find the local minimum of this function
- c) Write a Julia code that uses standard gradient descent technique to find the local minimum of $f(x)$. Check with your answer in (b).